An Independent Platform for Voice-Native OS Navigation by Individuals with Severe Motor Impairment

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Introduction

- A real-time voice agent closes a continuous perception → action loop: it hears speech, reasons over intent, and executes OS-level actions directly on a device (not just returning text like a chatbot).

- Audio is captured continuously, mapped to intent (ideally via a native audio-understanding model rather than a slower STT → LLM → TTS cascade), turned into executable commands, and run at the full OS level

Introduction

- After each action, the agent re-reads screen state (via screenshot/vision model or accessibility tree) and self-corrects on mismatch, since there's no human in the loop to catch errors.

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- Native audio-understanding models are preferred over the cascaded STT → LLM → TTS pipeline, since each stage in a cascade adds latency that compounds end-to-end

Existing Techniques

- Command-based: fast but rigid, no unstructured speech
- Browser agents: web-only, no OS/native access
- Cascaded STT→LLM→TTS: latency compounds per stage
- Intent classifiers: text-only, not real-time OS actions
- Voice assistants: edge = fixed intents; cloud = needs connectivity, English-first.

LITERATURE SURVEY

1. An embedded end-to-end voice assistant

PROBLEM

- Cloud assistants require Internet access increasing privacy risks
- Limited Offline Performance for existing assistants
- No End-to-End Architecture on prior works

SOLUTION

- Fully Offline Voice Assistant
- End-to-End Deep Learning Pipeline
- Transfer Learning for Italian ASR
- Configurable System Architecture
- Extensible to Multiple Languages
- Cloud-Comparable ASR Performance
- Reduced Response Latency
- Optimized for Embedded Devices

[1] Luca Lazzaroni, Francesco Bellotti, and Riccardo Berta. An embedded end-to-end voice assistant. Engineering Applications of Artificial Intelligence, 136:108998, 2024.

1. An embedded end-to-end voice assistant

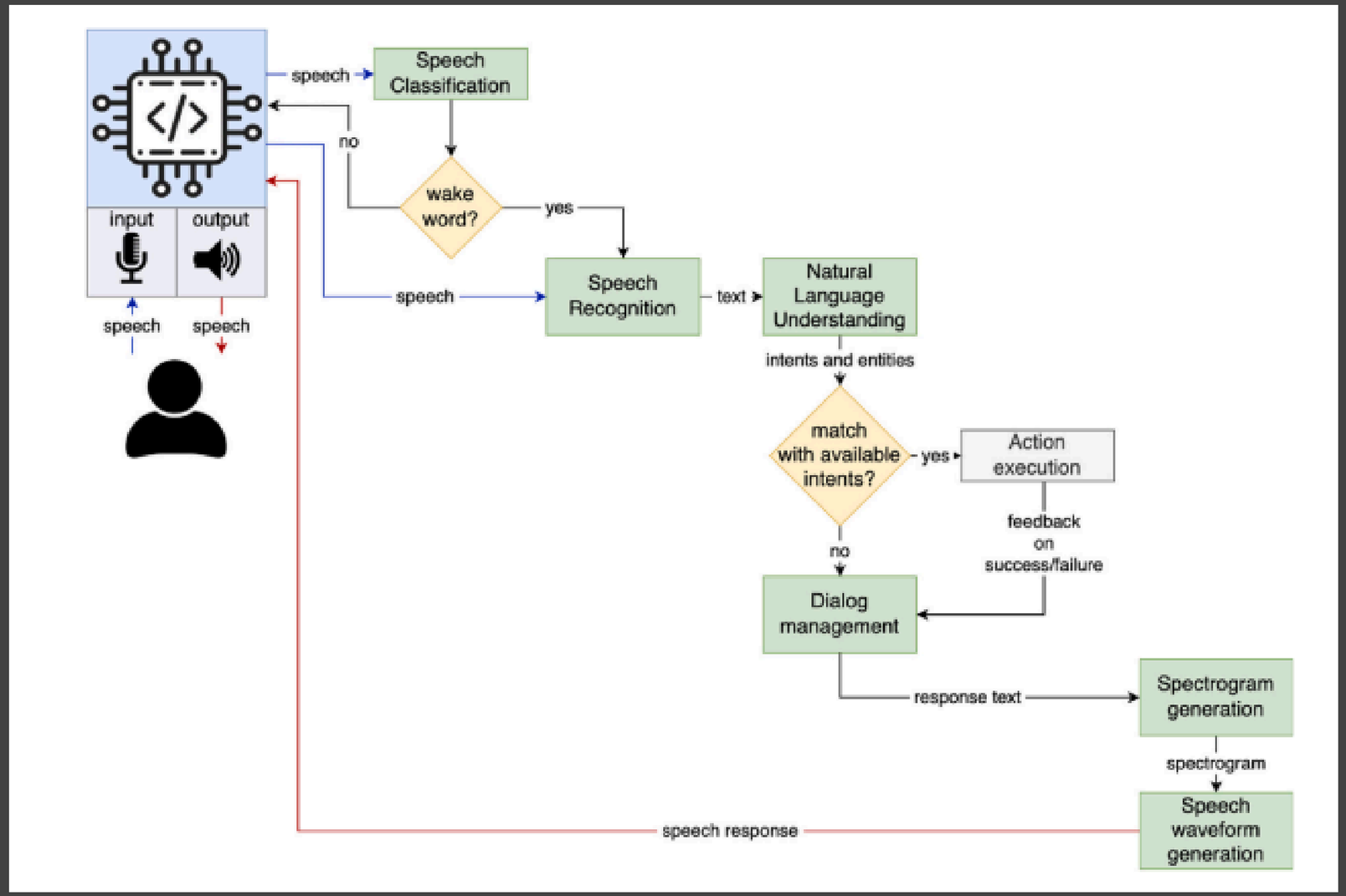
METHODOLOGY

1. Phase 1: Speech classification
2. Phase 2: Speech recognition
3. Phase 3: Language understanding
4. Phase 4: Speech synthesis
5. Phase 5: System deployment

[1] Luca Lazzaroni, Francesco Bellotti, and Riccardo Berta. An embedded end-to-end voice assistant. Engineering Applications of Artificial Intelligence, 136:108998, 2024.

1. An embedded end-to-end voice assistant

SYSTEM ARCHITECTURE



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1. An embedded end-to-end voice assistant

Results

Table 7
Error rates in noiseless and noisy environments.

Error type	Noiseless env	Noisy env
WER	17.71%	21.30%
CER	3.7%	5.1%
ICER	9.02%	9.35%
SER	3.09%	7.04%
IRER	9.77%	13.08%

[1] Luca Lazzaroni, Francesco Bellotti, and Riccardo Berta. An embedded end-to-end voice assistant. Engineering Applications of Artificial Intelligence, 136:108998, 2024.

2. Sink or SWIM: Tackling Real-Time ASR at Scale

PROBLEM

- Limited Concurrent Transcription
- High GPU Resource Usage
- Poor Multi-Client Scalability
- Limited Asynchronous Support
- Latency–Accuracy Trade-off

SOLUTION

- Multi-stream ASR inference
- Concurrent client support
- Buffer merging optimization
- Low-latency multilingual transcription

[2] Federico Bruzzone, Walter Cazzola, Matteo Brancaleoni, and Dario Pellegrino. Sink or SWIM: Tackling real-time ASR at scale. IEEE Transactions on Audio, Speech and Language Processing, 34:2238–2248, 2026.

2. Sink or SWIM: Tackling Real-Time ASR at Scale

METHODOLOGY

1. Phase 1 – Data Collection
2. Phase 2 – Intent Extraction
3. Phase 3 – Semantic Clustering
4. Phase 4 – Coverage Maximization
5. Phase 5 – Intent Prioritization

[2] Federico Bruzzone, Walter Cazzola, Matteo Brancaleoni, and Dario Pellegrino. Sink or SWIM: Tackling real-time ASR at scale. IEEE Transactions on Audio, Speech and Language Processing, 34:2238–2248, 2026.

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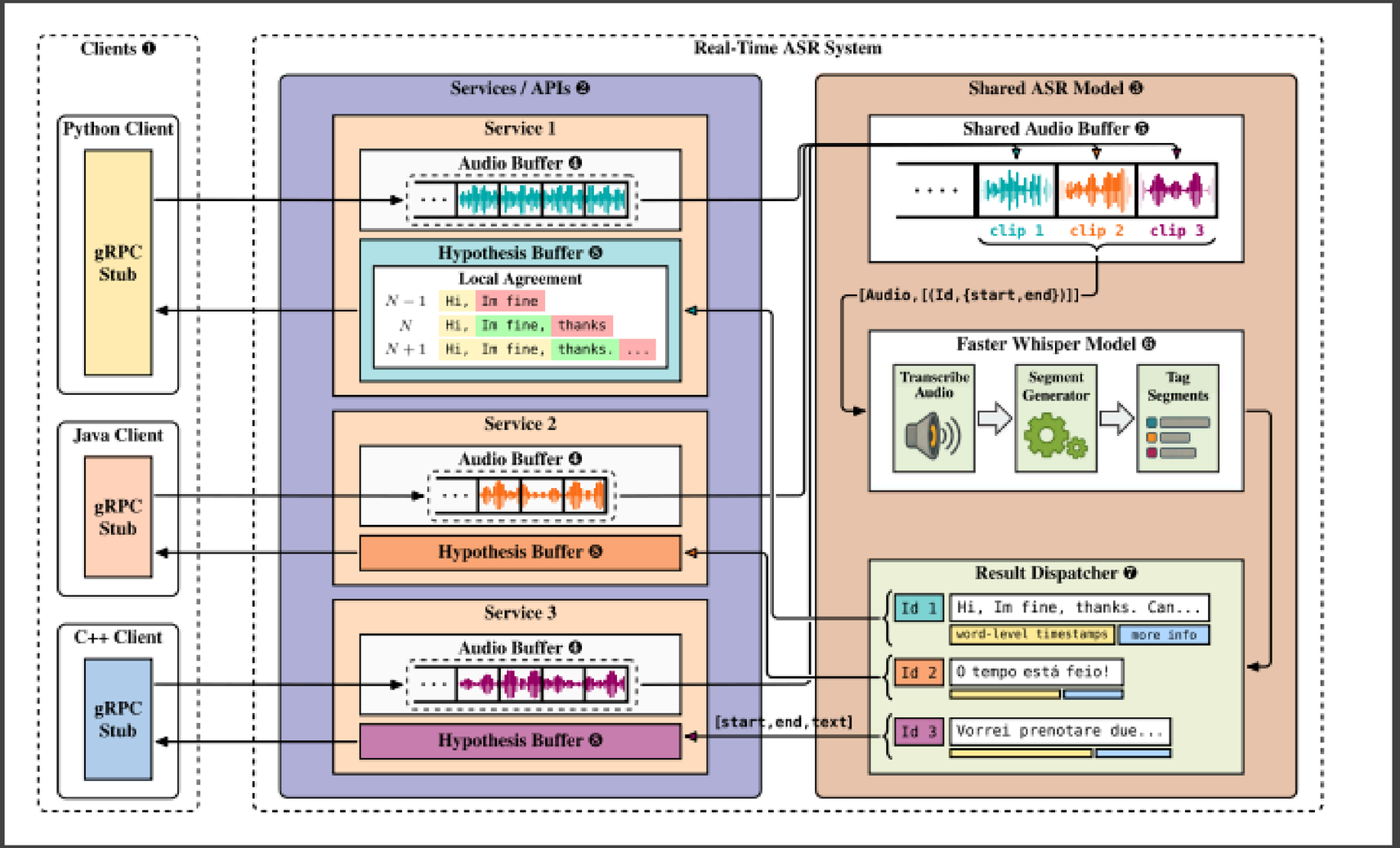

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2. Sink or SWIM: Tackling Real-Time ASR at Scale

Results

Model (Italian, Common Voice test set)	WER (%)	CER (%)	Transcription time (s)
Ours (QuartzNet15x5, NeMo-based)	117	312	215
Vosk	298	125	464
DeepSpeech (fine-tuned)	458	1.324	1.778

[2] Federico Bruzzone, Walter Cazzola, Matteo Brancaleoni, and Dario Pellegrino. Sink or SWIM: Tackling real-time ASR at scale. IEEE Transactions on Audio, Speech and Language Processing, 34:2238–2248, 2026.

3 . An intent recognition pipeline for conversational AI

PROBLEM

- Manual intent annotation is expensive
- Annotator bias affects quality
- No standard intent discovery method
- Lack of intent prioritization strategies
- Paraphrases and misspellings reduce coverage

SOLUTION

- Automated Intent Discovery
- Semantic Intent Expansion
- Misspelling & Synonym Detection
- Coverage-Based Intent Prioritization

[3] C. B. Chandrakala, R. Bhardwaj, and C. Pujari. An intent recognition pipeline for conversational AI. International Journal of Information Technology, 16:731–743, 2024.

3. An intent recognition pipeline for conversational AI

METHODOLOGY

1. Phase 1: Data Collection & Text Mining
2. Phase 2: Token-Based Intent Extraction
3. Phase 3: Semantic-Based Intent Extraction
4. Phase 4: Coverage Maximization
5. Phase 5: Intent Priority Ranking

[3] C. B. Chandrakala, R. Bhardwaj, and C. Pujari. An intent recognition pipeline for conversational AI. International Journal of Information Technology, 16:731–743, 2024.

3. An intent recognition pipeline for conversational AI

ALGORITHM

Input
$data_{corpus}$
Output
$Training_{set} = Max_{coverage}(data_{corpus})$
Steps
for do $word\ in\ data_{corpus}$
  if $word$=={stopword,number,punctuation} **then**
    $data_{set} = data_{corpus} - word$
  end if
end for
Token based intent extraction:
$\{unigram, , bigram, , program, tetragram\} \leftarrow n - gram(data_{set})$
$WordCount(data_{set})$
Semantic intent extraction:
$J(A, B) \leftarrow Jaccard\ Score$
$J(A, B) = \frac{\|A \cap B\|}{\|A \cup B\|} = \frac{\|A \cap B\|}{\|A\| + \|B\| - \|A \cap B\|}$
$where\ 0 \leq J \leq 1,\ '1'\ represents\ more\ similarity$
$A, B\ represents\ strings$
$b \leftarrow mean(nearest_{cluster})$
$a \leftarrow mean(intraDistance_{cluster}$
Silhouette Score$(a, b) = \frac{b-a}{max(a,b)}$
KMean$(data_{set}, Sentence_{embed})$
LDA$(data_{set}, threshold > 0.5)$
$\beta \leftarrow Extract_{Synonyms}(Intents)$
$\gamma \leftarrow Extract_{Misspellings}(Intents)$
$Training_{set} \leftarrow Max_{coverage}(\beta, \gamma)$

[3] C. B. Chandrakala, R. Bhardwaj, and C. Pujari. An intent recognition pipeline for conversational AI. International Journal of Information Technology, 16:731–743, 2024.

3. An intent recognition pipeline for conversational AI

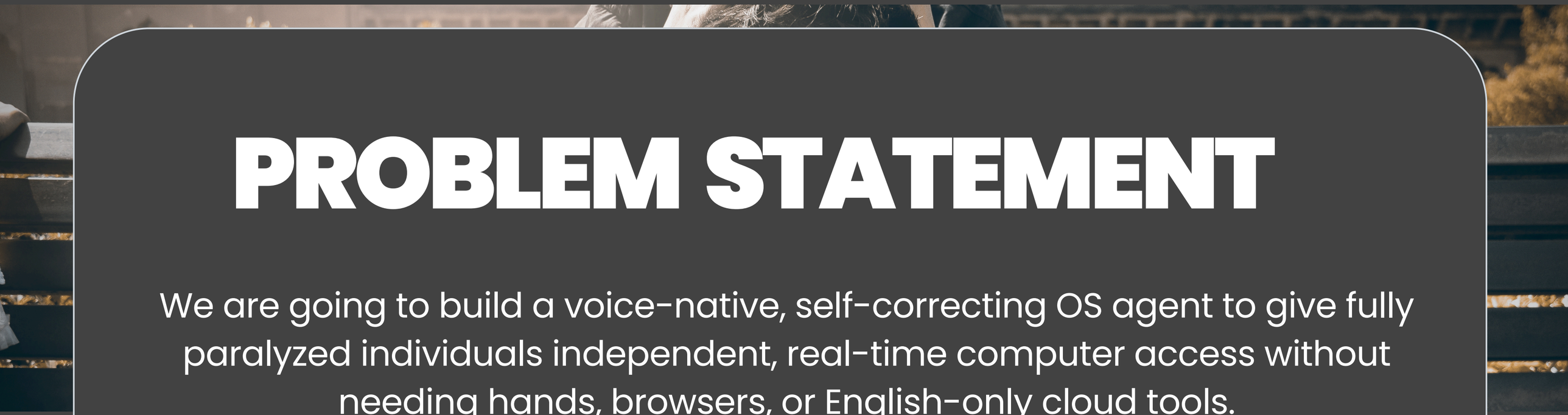
Results

Metric	Result
Optimal Clusters	**10** well-separated intent clusters identified using the **Silhouette Score** and **Elbow Method**.
Best Topic Model	**LSA** achieved the highest **Silhouette Score of 0.46** after grid search.
Coverage Improvement	Synonym (**Word2Vec**) and misspelling (**FastText**) expansion increased intent coverage by **10–11%**.
Intent Priority	**Phone Freeze (1.146) > Battery Problem (0.878) > Button Keywords (0.079)** with minimal overlap/conflict.
Pipeline Robustness	Multi-stage approach recovered missed intents across components, improving overall reliability.

[3] C. B. Chandrakala, R. Bhardwaj, and C. Pujari. An intent recognition pipeline for conversational AI. International Journal of Information Technology, 16:731–743, 2024.

RESEARCH GAP

Paper	Research Gap	Our Solution
Paper 1	LLMs omitted due to embedded device computational constraints.	Integrated LLM for conversation
Paper 2	No speech enhancement for noisy environment robustness.	AI Voice Isolation
Paper 3	Not evaluated in real-time accessibility-critical assistive environments.	Real time intent recognition for OS navigation



PROBLEM STATEMENT

We are going to build a voice-native, self-correcting OS agent to give fully paralyzed individuals independent, real-time computer access without needing hands, browsers, or English-only cloud tools.

OBJECTIVES

1. **Develop a hybrid agent architecture capable of operating in both offline (local LLM) and online modes, achieving an average query inference time of ≤15 seconds across both modes.**

2. **Implement a model-switching layer that allows users to select and swap between LLMs — including domestic models such as Sarvam AI — to expand support for Indian regional languages beyond English-only interaction.**

3. **Improve agent response efficiency by implementing a caching mechanism for frequently used user workflows, reducing repeated-task execution time by a measurable margin compared to uncached execution.**

DESIGN

load appropriate skill file

Code requirements lang, os, permissions

Generate Code

Destructive, permissions, guardrails

Code execute

single responsibility principle

METHODOLOGY

- **Phase 1 -** Speech Understanding & Intent Capture
- **Phase 2 -** Action Planning & Code Generation
- **Phase 3 -** Screen State Perception
- **Phase 4 -** Verification & Self-Correction
- **Phase 5 -** Latency Optimization & Offline Deployment

CONCLUSION

- This system gives fully paralyzed individuals independent, full-OS control via real-time, verified, self-correcting speech-to-action — going beyond rigid command grammars with LLM code generation, and beyond blind execution with screen-state checks that self-correct without human fallback.

- It's also built for India and built to be free, with regional-language support (Sarvam AI) and offline mode. Future work: dysarthric speech adaptation, hybrid deployment, and hardware-aware model selection.

References

[1] Luca Lazzaroni, Francesco Bellotti, and Riccardo Berta. An embedded end-to-end voice assistant. Engineering Applications of Artificial Intelligence, 136:108998, 2024.

[2] Federico Bruzzone, Walter Cazzola, Matteo Brancaleoni, and Dario Pellegrino. Sink or SWIM: Tackling realtime ASR at scale. IEEE Transactions on Audio, Speech and Language Processing, 34:2238– 2248, 2026.

[3] C. B. Chandrakala, R. Bhardwaj, and C. Pujari. An intent recognition pipeline for conversational AI. International Journal of Information Technology, 16:731–743, 2024.

THANK YOU